SectraTEK

COURSE OVERVIEW

# Five Patterns, One Discipline



A hands-on course in agentic AI for digital forensics. Five AI-agent patterns, then five labs that put AI to work on forensic evidence. Ten labs, all free in your browser.

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YOUR INSTRUCTOR

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## Mobile Threat Research Lead

SectraTEK Lab

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

**Mobile threat research**
Spyware, stalkerware and device tampering.

**AI for investigators**
Machine learning that speeds up the search.

**Teaching**
AI for Cybersecurity and Digital Forensics.

START HERE

# Who this course is for

You are learning cybersecurity or digital forensics, and you may be new to programming. This course teaches you to use AI agents as careful investigative assistants, step by step.

### No deep coding needed

You read and run notebooks, not write them from scratch. Every step says its purpose, input, and output in plain language.

### Realistic forensic cases

Synthetic mobile cases, an Android image, a bug report, an incident drill, and a real public memory image.

### One habit throughout

Every lab trains the same skill: say only what the evidence supports, and name what it does not.

THE CORE IDEA

# From a chatbot to an agent

## A plain AI answer

You ask once, it answers once. If the answer overreaches, nothing catches it.

One prompt, one reply.

## An AI agent

The model works inside a loop with a goal and a role. It can plan, use tools, check itself, and improve before it answers.

**A process, not a single reply.**

"Agentic AI" just means wrapping a model in a disciplined process. This course teaches five of those processes.

THE MAP OF THE COURSE

# The five agentic patterns



## Reflection

Draft, critique, revise.

LAB 01



## Tool Use

Call real tools to read the evidence.

LAB 02



## ReAct

Reason a step, act, observe, repeat.

LAB 03



## Planning

Break a hard case into ordered steps.

LAB 04



## Multiagent

Split roles, then reconcile findings.

LAB 05

**They build on each other.** A Multiagent team is made of ReAct agents that use Tools. Start at Lab 01 and go in order.

THE HABIT THAT MATTERS MOST

# Evidence-bounded reasoning

Every lab trains one discipline: separate what the evidence records from what you infer, and never claim more than it supports.



## Observation

What an artifact directly records.



## Inference

A reasonable interpretation of observations.



## Overclaim

A conclusion the evidence does not establish.

THE RECURRING EXAMPLE

A file upload that STARTED is not an upload that COMPLETED. An attempt is not a delivery. Catching that gap, in case after case, is what the whole course is really teaching.

WHAT EVERY LAB LOOKS LIKE

# See it, then reuse it

### Group activity

Optional: examine the raw evidence as a team and record a human baseline first.

### Build it by hand

Run the pattern one step at a time, so every model call is visible.

### Run the agent

Do the same workflow through a reusable, packaged class, to see the abstraction.

### Evidence check

Answer a few questions that tie each claim back to an artifact.

First you watch the loop work. Then you reuse it. Then you judge it. The judgment stays yours.

NO SETUP, NO COST

# Everything runs in your browser

## Google Colab

Each lab has an “Open in Colab” button. It opens in your browser. Nothing is installed on your computer.

## A free model key

One free Groq key works for all ten labs. It stays hidden as you paste it and is never saved.

## Self-contained notebooks

Each notebook writes its own case files and carries its own code, so you only ever open one file.

**WHAT YOU NEED**

A web browser and a free Groq account. No GPU, no admin rights, no Python install.

THE SYLLABUS, PART 1

# Labs 01-05: five patterns, five cases

| | | |
|---|---|---|
| 01 | Reflection | Does a leaked customer file prove exfiltration? |
| 02 | Tool Use | Was a stolen SUV photographed and listed for sale? |
| 03 | ReAct | Did a Signal image send while a phone was unattended? |
| 04 | Planning | Reconstruct a phone's timeline during a missing window. |
| 05 | Multiagent | Was a patient file transmitted, and is the custody sound? |

THE SYLLABUS, PART 2

# Labs 06-10: AI on forensic evidence

| | | |
|---|---|---|
| 06 | RAG over case reports | Answer questions from past case reports, citing the report used. |
| 07 | Agent over an Android image | Walk a phone image with read-only tools and cite the exact file. |
| 08 | Bug-report chatbot | Ask questions of an Android bug report, even one from your own phone. |
| 09 | Incident-response copilot | Work an incident from the IR plan, playbooks, and ISO 27001 ISMS. |
| 10 | Memory forensics | Volatility 3 plugins as tools, on a real public memory image. |

LABS 07-10

# The code checks the AI, too

Giving an AI access to the evidence limits what it can look at, not what it can say. So before you see an answer, the code checks it against what the tools really returned.

### Did it look?

At least one tool must have run for the question.

### Is every detail real?

Every file, IP address, process ID, and memory address must appear in what a tool returned.

### Is the citation honest?

The answer cannot cite a tool that never ran.

**IF IT FAILS**

The answer goes back to the AI to check or drop the claim. After two tries it arrives marked UNVERIFIED. The check catches invented details, not a wrong reading of real evidence. That part is still your job.

YOUR FIRST FIVE MINUTES

# How to begin

| | | |
|---|---|---|
| 1 | **Open the repository** | Go to the course repo and open the Lab 01 folder. |
| 2 | **Read the lab README** | It sets up the pattern and the case in plain language. |
| 3 | **Click Open in Colab** | The notebook opens in your browser, ready to run. |
| 4 | **Paste a free Groq key** | Create one at console.groq.com, paste when asked. |
| 5 | **Run it top to bottom** | Each step tells you its Purpose, Input, How it works, and Output. |

THE ONE RULE TO REMEMBER

# The AI assists. You decide.



## It speeds the work

Agents draft, search, and self-check far faster than a person can.



## It flags, it does not rule

An agent points at risk and evidence. It does not establish guilt.



## A human owns the verdict

A qualified examiner reads the evidence and signs the conclusion, always.

THE HONEST FRAMING

These patterns make an AI a better assistant to an examiner. Not one of them makes the AI the examiner.

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# Learn the patterns. Keep the discipline.

By the end you will have built all five agent patterns and used them on five kinds of forensic evidence. Each one a better assistant to an examiner, and never the examiner.

